RITIKA POOJARY

B.E. – CSE(Internet of Things & Cyber Security including Blockchain Technology)

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SUMMARY

B.E. CSE (IoT) graduate with an interest in software development and AI/ML. Familiar with Python, JavaScript, SQL, HTML, and CSS, with exposure to AI/ML through academic and personal projects. Willing to learn, adapt, and contribute to real-world projects.

SKILLS

- Programming Languages: Python
- UI Development: HTML, CSS, JavaScript
- Database: SQL
- Development Tool: Visual Studio Code, Git, GitHub , RoboFlow

EDUCATION

SIES Graduate School of Technology | B.E. - CSE | CGPA: 8.50 / 10 — **2025**
JVM New English School & Jr. College | 12th | MSBSHSE | Percentage: 86.67 / 100 — **2021**

PUBLICATION

30 April, 2025

SAFEConnect: ADual-Function Emergency Alert System Using Arduino Nano, GPS, and GSM Modules

Authors: Ritika Poojary, Dhanashree Nallakukkala, Pratiksha Kirdat, Nidhi Sonawane, Prof. Swapnil Wani Published at International Journal of Scientific Development and Research, Volume 10

PROJECTS

Civic Issue Detection

Jun 2026 – Jul 2026

Python | PyTorch| ResNet18 | CBAM | OpenCV | Flask | Render

- Developed an 8-class civic issue image classification model using a pretrained ResNet18 with transfer learning, enhanced with CBAM attention to focus on relevant visual features.
- Created a custom dataset by manually collecting and organizing images from the internet, applying preprocessing, augmentation, and oversampling to improve class balance and model generalization.
- Improved real-world prediction reliability using a multi-zoom inference approach with majority voting and confidence-based tie handling, addressing incorrect predictions caused by small or off-center objects and complex backgrounds.
- Evaluated the model on real-world images, integrated it with a Flask web application, and deployed the application on Render for remote access.

Medical Emergency Detection System

Jan 2026 – Feb 2026

Python| PyTorch | X3D-S | OpenCV | Flask | HTML | CSS | JavaScript

- Developed a real-time fall detection system using the X3D-S video classification model, trained on the Le2i and UR Fall Detection datasets to classify video sequences as Fall or Normal.
- Integrated YOLOv8 Pose to detect body keypoints and analyze movement after a detected fall, using changes in keypoint positions to estimate whether the person remained still.
- Implemented a fall-to-emergency verification pipeline where a detected fall triggers a timer, and persistent stillness for the configured duration escalates the system status from Fall Detected to Emergency, helping reduce immediate false alerts.
- Tested the trained model with a live camera feed and integrated the detection pipeline into a real-time web interface displaying fall status, pose/stillness information, timer progression, and emergency status.

CERTIFICATIONS

- Web Development — **20 Feb, 2024**
- Python Fundamentals for Beginners — **13 Jun, 2024**